(iii) $\sqrt{x}+\frac{1}{\sqrt[3]{x}}+e^x$

(iv) $\frac{3+2x-x^2}{x}$

(v) $x^3\,e^x$

(vi) $(x^2-3x+2)(x+1)$

(vii) $x^4-3\sin x+\cos x$

(viii) $\left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)^2$

2. Differentiate the following with respect to x.

(i) $\frac{e^x}{1+x}$ (ii) $\frac{x^2+x+1}{x^2-x+1}$

(iii) $\frac{e^x}{1+e^x}$

3. Differentiate the following with respect to x.

(i) $x\sin x$ (ii) $e^x\sin x$

(iii) $e^x(x+\log x)$ (iv) $\sin x\cos x$

(v) $x^3\,e^x$

4. Differentiate the following with respect to x.

(i) $\sin^2 x$ (ii) $\cos^2 x$

(iii) $\cos^3 x$ (iv) $\sqrt{1+x^2}$

(v) $(ax^2+bx+c)^n$ (vi) $\sin(x^2)$

(vii) $\frac{1}{\sqrt{1+x^2}}$

5.3.3 Derivative of implicit functions

For the implicit function $f(x,y) = 0$, differentiate each term with respect to x treating y as a function of x and then collect the terms of $\frac{dy}{dx}$ together on left hand side and remaining terms on the right hand side and then find $\frac{dy}{dx}$.

NOTE

If the function $f(x, y) = 0$ is an implicit function, then $\frac{dy}{dx}$ contains both x and y.

Example 5.36

If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ then find $\frac{dy}{dx}$.

Solution

$$ax^2+2hxy+by^2+2gx+2fy+c=0$$

Differentiating both side with respect to x,

$$2ax+2h\left(x\frac{dy}{dx}+y\right)+2by\frac{dy}{dx}+2g+2f\frac{dy}{dx}=0$$

$$(2ax+2hy+2g)+(2hx+2by+2f)\frac{dy}{dx}=0$$

$$\frac{dy}{dx}=-\frac{(ax+hy+g)}{(hx+by+f)}.$$

Example 5.37

If $x^3+y^3=3axy$, then find $\frac{dy}{dx}$.

Solution

$$x^3+y^3=3axy$$

Differentiating both sides with respect to x,

$$\frac{d}{dx}(x^3+y^3) = \frac{d}{dx}(3axy)$$

$$3x^2+3y^2\frac{dy}{dx} = 3a\left[x\frac{dy}{dx}+y\right]$$

$$x^2+y^2\frac{dy}{dx} = ax\frac{dy}{dx}+ay$$

$$(y^2-ax)\frac{dy}{dx} = (ay-x^2)$$

$$\frac{dy}{dx} = \frac{ay-x^2}{y^2-ax}$$

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